620 Stochastic Modelling in Finance

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September 2024

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1 Analytical Option Formulae

This part explores the pricing of European options, including vanilla call/put options and two types of digital options (cash-or-nothing and asset-or-nothing). Four models were applied and implemented: Black-Scholes, Bachelier, Black, and Displaced-Diffusion models.

1.1 Model Analysis

The Black-Scholes model assumes that the underlying asset price follows a log-normal distribution. Key formulas for vanilla options are:

$$d_1 = \frac{\ln(S/K) + \left(r + 0.5\sigma^2\right) T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T} \tag{1}$$

The Bachelier model assumes normal price changes and is suitable for low-volatility environments. The option price depends on the standard normal CDF:

$$C = (S-K)N\left(\frac{S-K}{\sigma\sqrt{T}}\right) + \sigma\sqrt{T}n\left(\frac{S-K}{\sigma\sqrt{T}}\right) \tag{2}$$

The Black model replaces the spot price S with the forward price F in the Black-Scholes framework, making it effective for interest rate derivatives:

$$C = e^{-rT}\left(FN(d_1) - KN(d_2)\right) \tag{3}$$

The Displaced-Diffusion model introduces a displacement factor , which adjusts the asset dynamics to handle skewed volatilities. The modified price is S=S+, and the structure resembles Black-Scholes.

1.2 Results

Vanilla options priced under the Black-Scholes model were stable and aligned with expectations in moderate volatility. The Bachelier model performed well in low-volatility scenarios but deviated under higher volatility. Digital options, particularly cash-or-nothing, showed high sensitivity to volatility and interest rates. The Displaced-Diffusion model offered greater flexibility for asymmetric volatility, while the Black model accurately captured characteristics of interest rate derivatives.

1.3 Conclusion

The Black-Scholes model is the standard for most equity options. However, the Bachelier model is better suited for low-volatility markets, and the Displaced-Diffusion model excels in skewed-volatility scenarios. Careful model selection based on market conditions is essential for accurate pricing.

2 Model Calibration

In this analysis, we study the option prices of the S&P500 (SPX) index and SPY as of December 1, 2020. The Implied Volatility Smile, reflecting the variation of implied volatility across strike prices, cannot be effectively captured by the traditional Black-Scholes model due to its limitations. Therefore, advanced models like the Displaced-Diffusion and SABR models are calibrated to fit the data. Key parameters such as β, σ, α, ρ, and ν are examined to evaluate their influence on the volatility smile, providing insights into the models' ability to describe market dynamics.

2.1 Data

2.1.1 Overview

We use SPX and SPY options datasets, which include option expiration times, strike prices, bid prices, ask prices, etc. The discount rate on this day is provided in the file `zero_rates_20201201.csv`.
Number of maturities: 3
Underlying asset prices on December 1, 2020:
SPX: 3662.45, SPY: 366.02.

2.1.2 Data Pre-processing

The following steps were applied for data pre-processing:

1. Standardized the strike prices.
2. Calculated the mid-price using the formula:

$$Mid - Price = \frac{BidPrice + AskPrice}{2}.$$

3. Selected option data based on expiration time and calculated the time to expiration T (in years).
4. Extracted the discount rate for each expiration time using interpolation.

2.2 The Displaced-Diffusion Model and SABR Model

2.2.1 Displaced-Diffusion Model

European Options (SPX)

- SPX option data was processed to extract the data for each expiration time.
- Tested model flexibility using different β values (0.8, 0.6, 0.4, 0.2).
- Fixed σ, and adjusted parameters by minimizing errors.

- **Key observations:**
  - $\beta = 0.8$: The volatility smile is smooth, the implied volatility curve is well-fitted, and the migration effect is noticeable.
  - $\beta = 0.6, 0.4, 0.2$: The smile curve becomes steeper, showing reduced sensitivity to the offset effect, but fitting errors for short-dated options increase.
- Smaller β values better capture the steepness of the smile curve but result in higher fitting errors, particularly for deep in-the-money or out-of-the-money options.

American Options (SPY)

- Same as the SPX step 1
- Applied the `DisplacedDiffusionVolatility` function to fit the model.
- Tested the effect of different β values.
- **Key observations:**
  - Results are similar to European options. However, due to the early exercise feature, the impact of β on the smile shape is more complex.
  - As β decreases, the symmetry of the smile diminishes significantly, and fitting quality for deep out-of-the-money options worsens.

2.2.2 SABR Model

European Options (SPX)

- Used SPX data with fixed $\beta = 0.7$, and calibrated the model to fit α, ρ, and ν.
- **Calibrated parameters:**
  - Short-term maturity: $\alpha = 1.212$, $\rho = -0.301$, $\nu = 5.460$.
  - Long-term maturity: $\alpha = 2.140$, $\rho = -0.575$, $\nu = 1.842$.
- ρ shows higher absolute values for short-term maturities, indicating pronounced asymmetry. ν decreases for long-term maturities, reflecting more stable market conditions.

American Options (SPY)

- Used SPY data, fixed $\beta = 0.7$, and calibrated the model using the `calibration` function.
- **Calibrated parameters:**
  - Short-term maturity: $\alpha = 0.665$, $\rho = -0.412$, $\nu = 5.250$.
  - Long-term maturity: $\alpha = 1.121$, $\rho = -0.633$, $\nu = 1.743$.
- For American options, ρ has higher absolute values, indicating more significant asymmetry. Short-term ν remains high, reflecting the market's sensitivity to uncertainty.

2.3 Conclusion

1. **Model Strengths and Weaknesses:**
   - **Displaced-Diffusion:** Simple and flexible but less effective for complex asymmetries in American options.
   - **SABR:** Handles asymmetry and steepness well, especially for short-term options, but is computationally intensive.
2. **European vs. American Options:**
   - American options introduce additional complexities due to early exercise features, impacting model calibration and fitting results.
3. **Future Improvements:**
   - Introduce advanced models like the Heston model for richer dynamics.
   - Use broader datasets to test robustness and consistency.

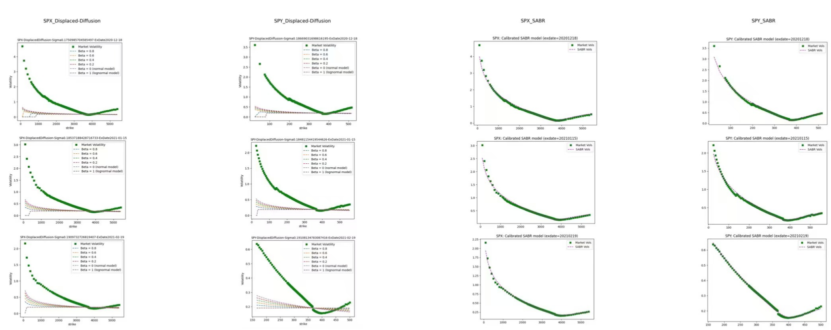

Figure 1: Total Plot of the Two Models

3 Static Replication

We evaluates SPX and SPY options using Black-Scholes, Bachelier, and SABR models. The analysis covers pricing and model-free integrated variance, derived under consistent assumptions: ATM volatility, 45-day maturity, and risk-free rate (r2). The aim is to identify model strengths and assess their suitability for different market conditions.

3.1 Data Preparation

Maturity (T)	Implied Volatility (σ_x)	Implied Volatility (σ_y)	Risk-Free Rate (r)
$T_1 = 18/12/2020$	0.1751	0.1867	0.465753
$T_2 = 15/01/2021$	0.1854	0.1848	0.123288
$T_3 = 19/02/2021$	0.1910	0.1911	0.219178

Table 1: Implied Volatilities (σ_x, σ_y) and Risk-Free Rates (r) for Different Maturities

3.2 Pricing

- Payoff Function

$$S_T^{1/3} + 1.5 \times \log(S_T) + 10.0 \tag{4}$$

Black-Scholes Model

$$S_T = S_0 \exp\left(\left(r - \frac{\sigma^2}{2}\right) T + \sigma Z\right), \quad Z \sim N(0,1) \tag{5}$$

Consequence: We calibrated the Black-Scholes model using the ATM option prices for SPX (strike: 3660) and SPY (strike: 366), resulting in σ values of 0.1854 and 0.1848, respectively. Based on these values, the derivative prices were calculated as 37.7012 for SPX and 25.9935 for SPY.

Bachelier Model

$$S_T = S_0 + \sigma W_T \tag{6}$$

Consequence: To determine the appropriate σ , we calibrated the Bachelier model to the ATM option prices for SPX (strike: 3660) and SPY (strike: 366). This resulted in σ, we calibrated the Black-Scholes model to the AT values of 0.1854 for SPX and 0.1848 for SPY. Using these, the derivative prices were calculated as 37.7231 for SPX and 26.0073 for SPY.

SABR Model

$$Price = e^{-rT} \cdot h(F) + \int_0^F \phi(K) P(F,K)\, dK + \int_F^\infty \phi(K) C(F,K)\, dK \tag{7}$$

$$F = S \cdot e^{rT} \tag{8}$$

Using the SABR parameters calibrated from the second question, we calculated the payoff-based prices for SPX and SPY options. Substituting the SABR parameters $(\alpha, \beta, \rho, \nu)$ into the payoff functions for SPX and SPY, the resulting prices were as follows:

Payoff Function		**SPX**	**SPY**
***Black-Scholes**	Price	37.701220	25.993484
	σ	0.185372	0.184812
***Bachelier**	Price	37.723135	26.007252
	σ	0.185372	0.184812
***SABR**	Price	37.669991	25.972923
	SABR params:		
	Alpha (α)	1.816001	0.90783421
	Beta (β)	0.7	0.7
	Rho (ρ)	-0.401415	-0.48729513
	Nu (ν)	2.791031	2.728857

3.3 "Model-free" integrated variance

"Model-free" integrated variance is a financial tool that does not rely on any specific model and is used to quantify the time accumulation of the variance (volatility) of the underlying asset.
- "Model-free" Integrated Variance :

$$\sigma_{MF}^2 T = E\left[\int_0^T \sigma_t^2 \, dt\right] \tag{9}$$

To compute the variance as described in the equation, follow these steps:
Put Option Integration: Calculate the integral of the put option price, $P(K)$, divided by K^2 over the range $[0, F]$, where F is a specific strike price cutoff.

$$\int_0^F \frac{P(K)}{K^2} \, dK$$

Call Option Integration: Similarly, compute the integral of the call option price, $C(K)$, divided by K^2 over the range $[F, \infty)$:

$$\int_F^\infty \frac{C(K)}{K^2} \, dK$$

Combine the Results: Add the two integrals obtained in the previous steps.
Adjust with the Exponential Term: Multiply the result of the combined integrals by e^{rT}, where r is the risk-free rate and T is the maturity, to account

for the time value of money:

$$2e^{rT}\left(\int_0^F \frac{P(K)}{K^2}\,dK + \int_F^\infty \frac{C(K)}{K^2}\,dK\right)$$

Interpret the Result: The result represents the expected integrated variance over the time period $[0, T]$.
This approach uses option prices to approximate the risk-neutral expected variance, assuming a no-arbitrage market model.

Model	**SPX Variance**	**SPY Variance**
Black-Scholes	0.004237	0.004496
Bachelier	0.004264	0.004238
SABR	0.006334	0.006013

3.3.1 Conclusion:

The SABR model estimates higher variances for SPX (0.006334) and SPY (0.006013), reflecting its sensitivity to volatility dynamics. In contrast, Black-Scholes (SPX: 0.004237, SPY: 0.004496) and Bachelier (SPX: 0.004264, SPY: 0.004238) yield lower variances, indicating limited flexibility for capturing market skew. This suggests SABR is better suited for environments with noticeable volatility asymmetry, while the other models may suffice in simpler scenarios.

4 Dynamic Hedging

4.1 The dynamic hedging strategy for an option

Suppose $S_0 = 100$, $\sigma = 0.2$, $r = 5\%$, $T = \frac{1}{12}$ year (i.e., 1 month), and $K = 100$. We use the Black-Scholes model to simulate the stock price. Suppose we sell this at-the-money call option, and we hedge N times during the life of the call option. Assume there are 21 trading days in the month.
The hedged portfolio is given by:

$$C_t = \phi_t S_t - \psi_t B_t, \tag{10}$$

where

$$\phi_t = \frac{\partial C}{\partial S} = \Phi\left(\frac{\ln\left(\frac{S_t}{K}\right) + \left(r + \frac{1}{2}\sigma^2\right)(T-t)}{\sigma\sqrt{T-t}}\right), \tag{11}$$

and

$$\psi_t B_t = Ke^{-r(T-t)}\Phi\left(\frac{\ln\left(\frac{S_t}{K}\right) + \left(r - \frac{1}{2}\sigma^2\right)(T-t)}{\sigma\sqrt{T-t}}\right). \tag{12}$$

4.2 The histograms of Hedging Error with N times

The discrete hedging strategy is simulated for 50,000 different, randomly generated scenarios of future stock price evolution. In each case, there are N = 21 and N = 84 hedging trades, which are evenly spaced over the life of the option.

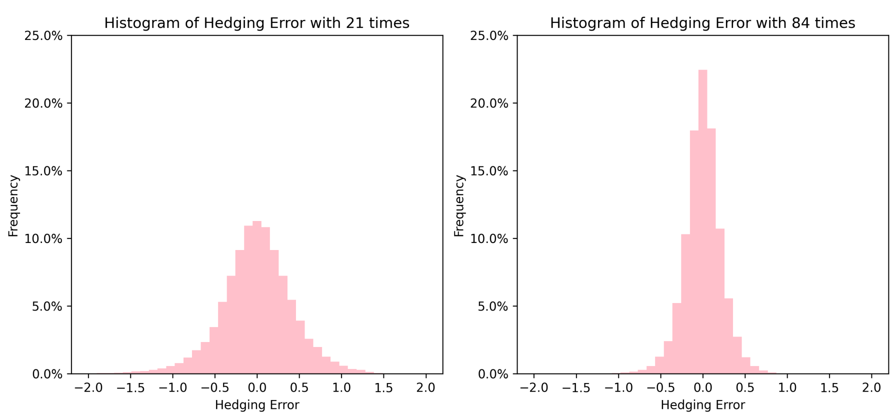

Figure 2: Hedging Error with N times

The histograms display the distribution of hedging errors under different hedging frequencies (21 and 84 times). Specifically:
- For $N = 21$, the histogram appears "short and wide," indicating that the error

distribution is more spread out with a larger standard deviation.
- For $N = 84$, the histogram is "tall and narrow," meaning the error distribution is more concentrated with a smaller standard deviation.
This difference can be attributed to the varying hedging frequencies. When the number of hedge adjustments is smaller (i.e., $N = 21$), the lower frequency makes it harder to capture the effects of market volatility, resulting in larger errors. On the other hand, when the hedging frequency is higher (i.e., $N = 84$), more frequent adjustments can respond more quickly to market movements, reducing the errors. Therefore, based on these two histograms, we can conclude that high-frequency hedging ($N = 84$) is more effective in reducing hedging errors.

4.3 Simulated profit/loss

The statistical summary of the simulated profit/loss is shown in the table below. Due to the randomness of Brownian Motion, the results of each run will be a little different. In the case of multiple runs, the largest change in value each time is the Mean Error.

Number of Trade	Mean Error	Standard Deviation Error	Std of Option Premium (%) Error
21	-0.002274	0.425235	16.927686
84	-0.000816	0.217273	8.649187

Table 2: Table showing the statistical summary of the simulated profit/loss

1. Average error: The average error reflects the degree of offset of the hedging strategy. A smaller average error (e.g. $N = 84$) means that during the hedging period, the value of the replicated portfolio is closer to the theoretical value at option expiration.
2. Standard deviation: Standard deviation reflects the volatility of the error. When $N = 21$, the larger the standard deviation, the greater the uncertainty and volatility caused by the low-frequency hedging strategy, and the larger the margin of error.
3. Standard deviation of option premium: When $N = 21$, the fluctuation of hedging error has a greater impact on option premium. At $N = 84$, volatility is reduced by about half, indicating that high-frequency hedging not only reduces hedging errors, but also makes the actual premium of options more stable.

4.4 Conclusion

From the above analysis, it can be seen that frequent hedging ($N = 84$) is more effective in reducing hedging errors than less hedging ($N = 21$). High-frequency hedging can adjust positions in a more timely manner, so as to follow market fluctuations more accurately and avoid the accumulation of large errors. In addition, the error volatility of high-frequency hedging is less, meaning that its performance is more stable.